



E-Bayesian Estimation for Exponential Distribution under Progressive Type II Censoring with DeGroot Loss Function

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Abstract:

This paper investigates E-Bayesian estimation for the parameter of the exponential distribution under the DeGroot loss function based on progressively Type-II censored data. We consider two scenarios: (i) when the sample size is fixed, and (ii) when the sample size is random. A simulation study verifies that a random sample size may lead to improvement in terms of estimated risk.

Keywords: DeGroot loss function, E-Bayesian estimation, Random sample size, Progressive Censoring, and Estimated risk.

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1 Introduction

The exponential distribution is known as an important lifetime distribution in modeling real-world phenomena. Let X follow an exponential distribution with parameter ξ . Then, we write $X \sim \text{Exp}(\xi)$, and the probability density function (PDF), and cumulative distribution function of X are $f(x; \xi) = \xi \exp(-\xi x)$, and $F(x; \xi) = 1 - \exp(-\xi x)$, respectively, where $x > 0$, and $\xi > 0$.

In reliability and life-testing experiments, censoring techniques are widely used to decrease experimental costs and shorten testing periods. Among these, progressive Type II censoring extends

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the classical Type II censoring by allowing the removal of surviving units during the experiment. Assume that n items are placed on test initially and the experiment is designed to observe m failures, where $1 \leq m \leq n$. After the first failure, R_1 surviving units are randomly withdrawn. Following the second failure, R_2 units are removed. This continues until the m th failure occurs, at which stage all remaining units are withdrawn, where $R_m = n - R_1 - R_2 - \dots - R_{m-1} - m$. The vector $R = (R_1, R_2, \dots, R_m)$ is the censoring scheme. The ordinary Type II censoring is obtained when $R = ((m-1)*0, n-m)$. For a detailed treatment of the theoretical foundations of progressive Type II censoring, see [Balakrishnan and Aggarwala \(2000\)](#).

In certain applications, maintaining a constant sample size is impossible due to the potential for missing observations for various reasons. [Srivastava \(1973\)](#) provided several applications when the sample size is random. A series of early studies, including [Raghunandan and Patil \(1972\)](#), [Buhrman \(1973\)](#), [Consul \(1984\)](#), and [Gupta and Gupta \(1984\)](#), laid the theoretical foundations for the distributions of order statistics in this setting. In a recent study, [Basiri and Asgharzadeh \(2021\)](#) addressed the problem of optimizing the random sample size in progressive Type-II censoring under a cost constraint.

A Bayesian solution to a statistical problem requires choosing a prior distribution and a loss function. In this paper, we focus on the DeGroot loss function (DLF), introduced by [DeGroot \(1970\)](#), defined as $L(\xi, \delta) = \left(\frac{\xi - \delta}{\delta}\right)^2$, where δ denotes an estimator of ξ . While some practitioners assume that a single prior can be elicited, prior information is often vague in practice, so any prior is only an approximation, see [Alhamaidah et al. \(2023\)](#). One method to overcome this difficulty is the E-Bayesian approach of [Han \(1997\)](#), see also [Alhamaidah et al. \(2023\)](#) and [Ghasabani et al. \(2026\)](#) for recent investigations.

In the remainder of this paper, we derive the E-Bayesian estimator for the exponential parameter under the DeGroot loss function based on progressively Type-II censored data. Two cases are examined: fixed and random sample sizes. A simulation study is also performed to compare and evaluate the estimators.

2 Estimation with a Fixed Sample Size ($N = n$)

Let $\mathbf{X}_{m:n} = (X_{1:m:n}, \dots, X_{m:m:n})$ denote a progressively Type II censored sample of size m from an exponential distribution with parameter ξ , where n is fixed. The likelihood function of ξ based

on the observed sample $\mathbf{x}_{m:n} = (x_1, \dots, x_m)$ is (see e.g. [Balakrishnan and Aggarwala \(2000\)](#))

$$\mathcal{L}(\xi | \mathbf{x}_{m:n}) = C_n \xi^m \exp \{-\xi t\}. \quad (2.1)$$

where $C_n = n(n - R_1 - 1)(n - R_1 - R_2 - 2) \cdots (n - R_1 - R_2 - \cdots - R_{m-1} - m + 1)$, and $t = \sum_{i=1}^m (R_i + 1)x_i$ is the observed value of $T = \sum_{i=1}^m (R_i + 1)X_{i:m:n}$. Maximizing the likelihood function with respect to ξ yields the maximum likelihood (ML) estimator of ξ as $\hat{\xi}_M = \frac{m}{T}$.

For Bayesian estimation, we consider the following gamma prior

$$\pi(\xi) = \frac{b^a}{\Gamma(a)} \xi^{a-1} e^{-b\xi}, \quad \xi > 0, \quad a > 0, b > 0. \quad (2.2)$$

Now, from (2.1) and (2.2), the posterior density of ξ given the data will become a gamma density with parameters $m + a$ and $b + t$, which is given as follows

$$\pi(\xi | \mathbf{x}_{m:n}) = \frac{(b + t)^{m+a}}{\Gamma(m + a)} \xi^{m+a-1} e^{-\xi(b+t)}.$$

Now, under the DLF, the Bayes estimator of ξ is obtained as

$$\tilde{\xi}_B = \frac{E(\xi^2 | \mathbf{X}_{m:n})}{E(\xi | \mathbf{X}_{m:n})} = \frac{m + a + 1}{b + T}.$$

2.1 E-Bayesian Estimation

Let $\tilde{\xi}_B$ be the Bayes estimator of ξ obtained under the prior distribution (2.2), and $\tilde{\pi}(a, b)$ represent the joint prior density of the hyperparameters a and b . Then, following [Han \(1997\)](#), the E-Bayesian estimator of ξ is defined as

$$\tilde{\xi}_{EB} = \int \int_{\mathbb{D}} \tilde{\xi}_B \tilde{\pi}(a, b) da db, \quad (2.3)$$

where $\mathbb{D}$ is the domain of a and b .

Following [Han \(1997\)](#), the hyperparameters a and b in the prior for ξ should be specified such that the prior density decreases in ξ . Differentiation of (2.2) shows this would be satisfied if $b > 0$ and $0 < a < 1$. However, larger values of b can reduce robustness, see [Berger \(1985\)](#); hence we impose $b \leq c$. We therefore adopt the joint prior for a and b as follows:

$$\tilde{\pi}(a, b) = \frac{1}{c}, \quad 0 < a < 1, \quad 0 < b < c. \quad (2.4)$$

Now, the E-Bayesian estimator of ξ under the DLF is obtained as

$$\begin{aligned} \tilde{\xi}_{EB} &= \int \int_{\mathbb{D}} \left[\frac{m + a + 1}{b + T} \right] \frac{1}{c} da db = \frac{1}{c} \int_0^1 (m + a + 1) da \int_0^c \frac{1}{b + T} db \\ &= \frac{2m + 3}{2c} \ln \left(\frac{c + T}{T} \right). \end{aligned}$$

3 Estimation with a Random Sample Size

In this section, we discuss parameter estimation when the sample size N is random. Let $\mathbf{X}_{m:N} = (X_{1:m:N}, \dots, X_{m:m:N})$ be progressively Type II censored order statistics from an exponential distribution with parameter ξ , where N is a discrete random variable supported on $\{m, m+1, \dots\}$. We assume N follows a truncated power series distribution starting at m with known parameter θ . Specifically, if N^* has a power series distribution with PMF

$$P_\theta(N^* = n) = \frac{h(n)\theta^n}{g(\theta)}, \quad n = 0, 1, \dots,$$

where $g(\theta) = \sum_{n=0}^{\infty} h(n)\theta^n$, then the PMF of N is

$$P_\theta(N = n) = \frac{h(n)\theta^n}{P_\theta(N^* \geq m)g(\theta)}, \quad n = m, m+1, \dots \quad (3.1)$$

Many well-known distributions such as the Poisson distribution belong to the class of power series distributions, see [Mahmoudi and Mahmoodian \(2015\)](#).

Let $\mathbf{x}_{m:N} = (x_1, \dots, x_m)$ be the observed sample. The joint likelihood of $X_{1:m:N}, \dots, X_{m:m:N}$ and N is ([Ghasabani et al. , 2026](#))

$$\mathcal{L}(\xi; \mathbf{x}_{m:N}, n) = \mathcal{L}(\xi, \mathbf{x}_{m:N} \mid N = n) P_\theta(N = n), \quad (3.2)$$

where $P_\theta(N = n)$ is given in (3.1) and

$$\mathcal{L}(\xi, \mathbf{x}_{m:N} \mid N = n) = C_n \xi^m e^{-\xi t^*}, \quad (3.3)$$

with t^* the observed value of $T^* = -\sum_{i=1}^m (R_i + 1)X_{i:m:N}$ and C_n as defined in Section 2. Substituting (3.1) and (3.3) into (3.2) yields

$$\mathcal{L}(\xi; \mathbf{x}_{m:N}, n) = \frac{h(n)\theta^n}{P_\theta(N^* \geq m)g(\theta)} C_n \xi^m e^{-\xi t^*}. \quad (3.4)$$

The ML estimator for ξ is $\hat{\xi}_M^* = m/T^*$, found by maximizing the likelihood function. From the prior (2.2) and the joint likelihood (3.4), the posterior density of ξ given $\mathbf{x}_{m:N}$ and $N = n$ is

$$\pi^*(\xi | \mathbf{x}_{m:N}, n) = \frac{(b + t^*)^{m+a}}{\Gamma(m+a)} \xi^{m+a-1} e^{-\xi(b+t^*)}.$$

Under the DLF, the Bayes estimator of ξ is given by

$$\tilde{\xi}_B^* = \frac{E(\xi^2 | \mathbf{X}_{m:N}, N)}{E(\xi | \mathbf{X}_{m:N}, N)} = \frac{m+a+1}{b+T^*}. \quad (3.5)$$

Now, using (2.3), (2.4) and (3.5), the E-Bayesian estimator of ξ under the DLF is obtained as

$$\tilde{\xi}_{EB}^* = \int \int_{\mathbb{D}} \left[\frac{m+a+1}{b+T^*} \right] \frac{1}{c} da db = \frac{2m+3}{2c} \ln \left(\frac{c+T^*}{T^*} \right).$$

4 Simulation Study

This section presents a simulation study comparing the proposed point estimators. We take $m = 10$ and 20, with N following a truncated Poisson distribution starting at m with parameter θ . Two scenarios for N are examined: (i) Fixed case: $N = 15$ for $m = 10$, and $N = 30$ for $m = 20$. (ii) Random case: N is generated from the truncated Poisson distribution starting at $n = 10, 20$ with parameter θ_0 , chosen such that $E_{\theta_0}(N) = 15$ and 30, respectively. For $E_{\theta_0}(N) = 15$, we have $\theta_0 = 14.33724$. Moreover, from [Ghasabani et al. \(2026\)](#), for $E_{\theta_0}(N) = 30$, we have $\theta_0 = 29.69595$. We assume that the censoring scheme is $((m-1)*0, N-m)$, which corresponds to ordinary Type II right censoring. We generate $X_{1:m:N}, \dots, X_{m:m:N}$ from $Exp(\xi)$ with $\xi = 0.5, 1$ and 2. For each replication, we compute ML, Bayesian, and E-Bayesian estimates under the DLF with $c = 2, 3, 5$. For Bayesian estimation, following [Ghasabani et al. \(2026\)](#), we choose $a = 0.5$, which is the prior mean of a , and $b = a/\xi$ so that $E(\xi|a, b) = a/b$ equals the true ξ . With $S = 10000$ replications, the estimated risk (ER), and the estimated bias (bias for short) of an estimator $\hat{\xi}$ are

$$ER(\hat{\xi}) = \frac{1}{S} \sum_{i=1}^S \left(\frac{\hat{\xi}_i - \xi}{\hat{\xi}_i} \right)^2, \quad \text{and} \quad bias(\hat{\xi}) = \frac{1}{S} \sum_{i=1}^S (\hat{\xi}_i - \xi),$$

respectively, where $\hat{\xi}_i$ is the estimate from the i th replication. Tables 1 reports the ERs and biases for all estimators.

From Table 1, we observe that estimators based on a random sample size produce lower ERs than those based on a fixed sample size for $m = 10$ when $\xi = 0.5$ and for $m = 20$. In terms of bias, the random sample size estimators outperform the fixed ones except for $\xi = 2$. Among the ML, Bayesian, and E-Bayesian estimators, the E-Bayesian estimators with $c = 5$ perform best for $\xi = 0.5$ in terms of ER, while those with $c = 3$ are best for $\xi = 1$ in this respect. Regarding bias, the E-Bayesian estimators with $c = 5$ achieve the best bias performance for $\xi = 0.5$ and 1. However, for $\xi = 2$, the Bayesian estimators provide the best performance in both ER and bias. Both ER and bias are decreasing with respect to m in most cases, as expected.

Table 1: ERs (first row) and biases (second row) of the ML, Bayesian and E-Bayesian estimators.

$m = 10$							
N		$\xi = 0.5$		$\xi = 1$		$\xi = 2$	
		Fixed	Random	Fixed	Random	Fixed	Random
ML		0.27500	0.27479	0.10125	0.10147	1.40744	1.41050
		0.61266	0.61221	0.11266	0.11221	-0.88734	-0.88779
Bayes		0.29107	0.29087	0.08403	0.08411	0.92038	0.92253
		0.63887	0.63828	0.20464	0.20406	-0.75918	-0.75974
E-Bayes	$c = 2$	0.29269	0.29248	0.07912	0.07924	1.13316	1.13555
		0.64385	0.64328	0.14385	0.14328	-0.85615	-0.85672
	$c = 3$	0.27220	0.27200	0.07753	0.07769	1.28758	1.29013
		0.58990	0.58935	0.08990	0.08935	-0.91010	-0.91065
	$c = 5$	0.23509	0.23491	0.08394	0.08417	1.62293	1.62579
		0.50032	0.49982	0.00032	-0.00018	-0.99968	-1.00018
$m = 20$							
N		$\xi = 0.5$		$\xi = 1$		$\xi = 2$	
		Fixed	Random	Fixed	Random	Fixed	Random
ML		0.26309	0.26273	0.05042	0.04890	1.19777	1.19153
		0.55401	0.55316	0.05401	0.05316	-0.94599	-0.94684
Bayes		0.27313	0.27283	0.04588	0.04457	0.95227	0.94687
		0.57351	0.57271	0.10244	0.10158	-0.88247	-0.88335
E-Bayes	$c = 2$	0.27355	0.27324	0.04432	0.04300	1.07752	1.07211
		0.57455	0.57374	0.07455	0.07374	-0.92545	-0.92626
	$c = 3$	0.26225	0.26194	0.04378	0.04246	1.16469	1.15928
		0.54828	0.54752	0.04828	0.04752	-0.95172	-0.95248
	$c = 5$	0.24087	0.24056	0.04569	0.04437	1.34716	1.34172
		0.50064	0.49996	0.00064	-0.00004	-0.99936	-1.00004

Discussion and Conclusion

In this paper, we explored E-Bayesian inference for the exponential distribution parameter under the DeGroot loss function, based on progressively Type-II censored data. The results were obtained for two sampling scenarios: fixed and random sample sizes. The simulation results indicated that estimators based on a random sample size may offer an advantage in terms of estimated risk over those based on a fixed sample size. All computations were performed using the R software ([R Core Team, 2024](#)), with packages `nleqslv` ([Hasselman, 2023](#)), and `extraDistr` package ([Wolodzko, 2023](#)).

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References

- Alhamaidah, A., Naghizadeh Qomi, M. and Kiapour, A. (2023), Comparison of E-Bayesian estimators in Burr XII model using E-PMSE based on record values, *Statistics, Optimization & Information Computing*, 11(3), 709-718.
- Balakrishnan, N. and Aggarwala, R. (2000), *Progressive censoring: Theory, methods, and applications*, Boston, Birkhäuser.
- Basiri, E., and Asgharzadeh, A. (2021), Optimal random sample size in progressively Type-II censoring based on cost constraint for the proportional hazards family, *Journal of Statistical Computation and Simulation*, 91(15), 3154-3169.
- Berger, J.O. (1985), *Statistical decision theory and Bayesian analysis*, 2nd edn., New York, Springer-Verlag.
- Buhrman, J.M. (1973), On order statistics when the sample size has a binomial distribution, *Statistica Neerlandica*, 27(3), 125-126.
- Consul, P.C. (1984), On the distributions of order statistics for a random sample size, *Statistica Neerlandica*, 38(4), 249-256.
- DeGroot, M.H. (1970), *Optimal statistical decisions*, McGraw-Hill Book Company.

- Ghasabani, R., MirMostafaei, S.M.T.K., Naghizadeh Qomi, M. and Basiri, E. (2026), E-Bayesian estimation in the proportional hazard rate family based on progressively Type II censored data with fixed and random sample sizes. *Journal of the Indian Society for Probability and Statistics*, 27(1), 61-107.
- Gupta D. and Gupta R.C. (1984), On the distribution of order statistics for a random sample size, *Statistica Neerlandica*, 38(1), 13-19.
- Han, M. (1997), The structure of hierarchical prior distribution and its applications, *Chinese Operations Research and Management Science*, 6(3), 31-40.
- Hasselmann, B. (2023), nleqslv: Solve systems of nonlinear equations, R package version 3.3.5, <https://CRAN.R-project.org/package=nleqslv>.
- Mahmoudi, E. and Mahmoodian, H. (2015), Normal power series class of distributions: Model, properties and applications, arXiv preprint, <https://arxiv.org/abs/1510.07180>.
- R Core Team (2024), R: A language and environment for statistical computing, R Foundation for Statistical Computing, Vienna, Austria.
- Raghunandanan, K. and Patil, S.A. (1972), On order statistics for random sample size, *Statistica Neerlandica*, 26(4), 121-126.
- Srivastava, R.C. (1973), Estimation of probability density function based on random number of observations with applications, *International Statistical Review / Revue Internationale de Statistique*, 41(1), 77-86.
- Wolodzko, T. (2023), extraDistr: Additional univariate and multivariate distributions, R package version 1.10.0, <https://CRAN.R-project.org/package=extraDistr>.